

## **Weekly Diary – Week 20**

**Internship Organization:** Vikaspedia

**Duration:** 19/05/2026 – 24/05/2026

### **Objective of the Week**

The objective of this week was to understand the architecture of the Indian Language Web Index project and begin implementing modules for collecting, processing, and indexing multilingual web content. The focus was on studying language-specific challenges and preparing the data processing pipeline.

#### **Date- 19/May**

Studied the project requirements and architecture of the Indian Language Web Index system. Understood how multilingual web content can be collected, processed, indexed, and made searchable. Researched challenges associated with Indian languages such as different scripts, encoding standards, Unicode handling, and language identification. Analyzed the importance of supporting multiple regional languages for improving accessibility and information retrieval.

#### **Date- 20/May**

Explored various Indian language datasets and web sources that could be used for indexing. Studied how multilingual content is represented on websites and analyzed differences in character encoding across languages. Conducted experiments with Hindi, Punjabi, Marathi, Tamil, and Telugu text to verify proper Unicode rendering and storage. Prepared sample datasets for testing indexing functionality.

#### **Day- 21/May**

Worked on designing the data collection and preprocessing workflow. Studied methods for extracting textual content from webpages while removing unwanted HTML elements and formatting tags. Implemented basic text-cleaning operations to remove special characters, duplicate spaces, and irrelevant content. Conducted testing with multilingual webpages to verify extraction accuracy.

#### **Date- 22/May**

Implemented language detection mechanisms to identify the language of extracted content automatically. Explored available language identification libraries and evaluated their performance on Indian language text samples. Conducted tests with content written in different scripts and documented the accuracy of language classification results.

**Date- 23/May**

Developed the initial indexing structure for storing processed content. Studied search indexing concepts and analyzed how multilingual text can be organized efficiently for retrieval operations. Implemented sample indexing functionality and verified that content from different languages could be stored and accessed correctly.

**Date- 24/May**

Performed testing of the data extraction and indexing workflow using multilingual sample datasets. Verified that the extracted content retained its original language characteristics after processing. Documented observations regarding indexing performance, language detection accuracy, and preprocessing effectiveness.

**Conclusion**

By the end of the week, a clear understanding of multilingual content processing was achieved. The data extraction, preprocessing, language detection, and basic indexing mechanisms were successfully implemented, creating the foundation for the Indian Language Web Index system.